PROGRAMME : Diploma Programme in ME / PS / EE / IF / CM / EL / AE / DD / MK
COURSE : Industrial Organization and Management(IOM) **COURSE CODE :** 232402

LEARNING AND ASSESSMENT SCHEME:

Learning Scheme					Credits	Assessment Scheme											Total Marks
Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL and TSL Practical				Based on self Learning		
CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SA		
							Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
03	00	00	01	04	02	03 Hrs	30	70	100	40	--	--	--	--	25	10	125

IKS Content : Total Learning Hours for Term : 01 Hrs

1.0 RATIONALE:

Diploma engineers are expected to work in industry. It is expected that they work efficiently and effectively with optimum utilization of available resources. The course on industrial management and organization can be a valuable addition to a technical diploma holder offering personal and professional growth opportunities.

2.0 INDUSTRY / EMPLOYER EXPECTED OUTCOME:

The aim of this course is to help the student to attain the following industry identified outcome through various teaching learning experiences: Blend technical expertise with management principles and contribute effectively to organizational goals.

3.0 COURSE OUTCOMES:

The course content should be taught and learning imparted in such a manner that students are able to acquire required learning outcome in cognitive, psychomotor and affective domain to demonstrate following course outcomes:

1. Use the basic management principles to execute work-place activities.
2. Choose a type of industrial ownership or organization suitable for relevant business.
3. Utilize suitable techniques of material management and work as a production supervisor.
4. Relate the functions of Human Resource and Marketing department.
5. Use the Project Management and Safety practices in industry.

4.0 COURSE DETAILS:

Unit-I Introduction to Management	1a. Justify the importance of management thoughts in Indian knowledge system 1b. Differentiate between Management, Administration and Organization. 1c. Explain the role and importance of management. 1d. Explain the concept and steps for implementation of scientific management. 1e. Describe various principles and functions of management. 1f. Identify appropriate level of management with skills. 1g. Recognize duties and responsibilities of a Supervisor.	1.1 Evolution of management thoughts from ancient /medieval to modern times in India (IKS) 1.2 Concept of Management, Administration, Organization. Relationship between Management, Administration and Organization. 1.3 Importance and Characteristics of Management, Managerial Skills 1.4 Scientific Management, Basic Approach of Scientific Management. 1.5 Principles of management (14 Principles of Henry Fayol), Functions of Management. 1.6 Levels of Management and their functions with skills at different levels 1.7 Duties and responsibilities of a Supervisor.	9
Unit-II Industrial Organization and Ownership	2a. Suggest the suitable type of organization structure for a given organization. 2b. Select a type of industrial ownership suitable for relevant business.	2.1 Types of organization structures as line, line and staff, functional organizational structures, their merits, demerits and applications. 2.2 Industrial Ownership: Concept, Advantages, Disadvantages, and applications for various types of Industrial ownership like Single Ownership, Partnership, Joint Stock Company, Cooperative Organization, Public Sector and private sectors.	7
Unit-III Production and Material Management	3a. Explain the types of production systems. 3b. Describe the Functions and objectives of Material Management. 3c. Comprehend the steps for purchase procedure. 3d. Describe the types and functions of inventory. 3e. Explain the necessity and procedure for ABC analysis. 3f. Describe the aim and ways for Total Quality Management.	**A. Production Management** 3.1 Introduction to Production System, Methods of Production job, batch and mass), Types of production systems (Intermittent and Continuous), Concept of One time Project System. **B. Material Management** 3.2 Introduction and definition of Material Management, Functions and objectives of Material Management 3.3 Objectives of Purchasing department, Purchasing Procedure. 3.4 Concept of Inventory and Inventory Control, Inventory classification, functions of inventories. 3.5 ABC Analysis - Necessity & Steps **C. Quality Management** 3.6 Quality Management Practices:	14

	3g. Explain various quality management Practices. 3h. State the importance of Supply Chain and Logistic Management. 3i. Describe the role of information technology in Supply Chain and Logistic Management.	Concept, advantages and limitations of: ISO 9000, Just-In-Time (JIT), Lean, Agile, Kaizen, Six Sigma, 5S techniques. 3.7 Overview of Supply Chain and Logistic Management. 3.8 Role of information technology, Supply Chain and Logistic Management	
Unit-IV Marketing and Human Resource Management	4a. Describe the goals and functions of marketing management. 4b. Explain the significance of marketing strategy. 4c. Illustrate the various aspects of event management. 4d. Comprehend the recruitment procedure for employee selection. 4e. Explain the importance and types of employee training.	**A. Marketing Management** 4.1 Introduction to marketing management, definition, concept and functions and goals of marketing management. Selling Versus Marketing Concept. 4.2 Marketing Strategy, market segmentation, marketing mix (4P's), Sales Promotion methods, International Marketing, Digital Marketing. 4.3 Event Management: types, different aspects of event management. **B. Human Resource Management** 4.4 Introduction, definition, concept and functions of Personnel Management. Recruitment and selection of employees (Steps and overview) 4.5 Training of human resources: objectives, advantages and methods of training.	9
Unit-V Safety and Project Management	5a. Explain the importance of safety management and preventive measure to avoid accidents. 5b. Recognize importance of discipline in industry. 5c. Recognize the fundamentals of project management. 5d. State the terms for network analysis.	**A. Safety Management** 5.1 Introduction to safety, Losses due to Accident, Causes of Accident in industries, Factors responsible for Accidents, Preventive Measures, General Precautions, Hazards. 5.2 Discipline in Industry. **C. Project Management** 5.3 Introduction to Project Management, Tools and Techniques for Project Management, Overview of Project life-cycle phases. Benefits from Project Management. 5.4 Introduction to Network analysis, Terms related to Network Planning Methods with a suitable example. (No numerical or problem expected)	6
		Total	**45**

5.0 **SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):**

Unit No.	Unit Title	Distribution of Theory Marks			
		R Level	U Level	A and above Levels	Total Marks
I	**Introduction to Management**	2	10	2	14
II	**Industrial Organization and Ownership**	2	4	4	10
III	**Production and Material Management**	6	12	4	22
IV	**Marketing and Human Resource Management**	6	6	2	14
V	**Safety and Project Management**	4	4	2	10
	TOTAL	**20**	**36**	**14**	**70**

6.0 **LABORATORY LEARNING OUTCOME AND ALLIED PRACTICAL/ TUTORIAL EXPERIENCES:**

Not Applicable

7.0 **SELF LEARNING: (Assignment / Activities for Specific Learning / Skill Development / Online Courses / Micro projects (Minimum One activity on each Unit)**

Other than the classroom learning, following are the **suggested** student-related co-curricular activities. This can be undertaken to accelerate the attainment of the various outcomes in the course. Each student must perform the all the compulsory activities and at least one activity in each group. Following are suggested as sample assignments. Faculty is expected to assign individual assignment to each student and keep record in hard/soft copy as required for assessment.

Compulsory Assignment:

- Prepare a plan to achieve professional goals by focus on self-management skills. Submit a report.
- Prepare a case study on management lessons based on life of a personality who has inspired you.
- Prepare a report on management thoughts from ancient times in India.

Group I:

1. Prepare a report or presentation on principles of management citing suitable examples.
2. Prepare a report or presentation on functions of management citing suitable examples.
3. Prepare a report or presentation on various approaches for scientific management.
4. Prepare a report or presentation on levels of management in an organization.
5. Prepare a report or presentation on organization structure in any industry.
6. Prepare a report or presentation on ownership in any industry.
7. Prepare a report or presentation on case study focussing on effect of good management practices in growth of industry.
8. Prepare a report or presentation on various functions done by supervisor.
9. Prepare a report or presentation on any one industrial act.
10. Prepare a report or presentation on factors of industrial disputes.

Group II:

1. Prepare a report or presentation on any one production system in industry.
2. Prepare a report or presentation on purchasing procedure for a given item.
3. Perform an ABC or EOQ analysis on inventory in any industry.
4. Present a seminar on any one modern technique used for material management.
5. Prepare a report or presentation used for marketing strategy for a new product launch.

6. Collect information and prepare a report on various types of employment tests used during selection process.
7. Collect information and prepare a report on criteria of physical examination for various job roles in selection process.
8. Prepare a report or presentation on training procedure
9. Prepare a report or presentation on type of training used for any one level of staff in industry.
10. Prepare a network diagram for your proposed final year project.
11. Prepare a report or presentation or video based on safety measures to be followed in specific industry.
12. Prepare a report or presentation on causes of accidents

8.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):

1. Discuss various case studies to demonstrate the principles and functions of management.
2. Conduct a role-play to demonstrate functions of management.
3. Arrange expert lecture of an Industry Person.
4. Encourage students to visit various websites of organization.
5. Encourage students to read a book on Management
6. Arrange an industrial visit to Manufacturing/Assembly unit.
7. Conduct an online quiz on managerial skills

9.0 ASSESSMENT METHODOLOGY

1. Formative assessment (Assessment for Learning) Rubrics for continuous assessment based on process and product related performance indicators (25 marks)
2. Two unit tests of 30 marks and best of two.
3. End of the term examination (70 marks)

a. LEARNING RESOURCES:

A) Books

Sr. No.	Title of Book	Author	Publication
1.	Industrial Engineering & Management	Dr. O. P. Khanna	Dhanpat Rai & Sons New Delhi
2.	Industrial Engineering & Management	Banga & Sharma	Khanna Publication
3.	Engineering Management	A. K. Gupta	S. Chand, ISBN: 812812-5,2007, 2nd edition
4.	Basic Managerial Skills for All	E. H. McGrath	PHI ISBN: 978-8120343146, 2011, 9th Edition
5.	Management Concepts and Cases	Andrew DuBrin	Cengage Learning, ISBN: 978-8131510537, 2009,9th edition
6.	Indian Management	Subhash Sharma	New Age International Private Limited ; ISBN- 978-9389802412, 2020, 1st edition

B) Software/Learning Websites

Sr.No	Link/Portal	Description
1	https://www.debonogroup.com/services/core-programs/six-thinking-hats/	Six Thinking Hats
2	https://hbr.org/1981/09/managing-human-resources	HR Management
3	https://theproductmanager.com/topics/agile-product-management	Agile Product Management
4	https://www.cdlogistics.ca/freight-news/the-5-components-of-supply-chain-management	Supply Chain Management

5	https://www.infosectrain.com/blog/understanding-the-concepts -of-gantt-chart-and-critical-path-methodology-cpm	PERT, CPM, GANTT Chart
6	https://www.simplilearn.com/best-management-tools-article	Management Tools
7	https://www.psychometrica.in/free-online-psychometric-tests.html	Psychometric Tests
8	https://www.investopedia.com/terms/e/erp.asp	ERP
9	https://asq.org/quality-resources/quality-management-system	QMS
10	https://testlify.com/test-library/creative-thinking/	Psychometric Tests
11	https://www.mindtools.com/	Management Skills
12	https://www.investopedia.com/terms/d/digital-marketing.asp	Digital Marketing
Note: Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students.		

C) Major Equipments/ Instruments with Broad Specifications

Not Applicable

10.0 **MAPPING MATRIX OF PO's, CO's and PSO's:**

Course Outcomes	Programme Outcomes (PO's)							Programme Specific Outcomes (PSO's)		
	1	**2**	**3**	**4**	**5**	**6**	**7**	**1**	**2**	**3**
CO1	L	L	L	--	L	M	H			
CO2	L	L	L	--	L	H	H			
CO3	L	H	H	--	L	H	H			
CO4	L	H	H	--	L	H	H			
CO5	L	M	M	--	M	H	H			

H: High Relationship, M: Moderate Relationship, L: Low Relationship.

11.0 **SUGGESTED QUESTION PAPER PROFILE**

UNIT No	CO	Marks per Unit	1.35 Times marks	Question Number Wise Marks						Actual Distribution of Marks
				01	**02**	**03**	**04**	**05**	**06**	
I	CO1	14	**19**	2	4	4	4	4		**18**
II	CO2	10	**13**	4	4				4	**12**
III	CO3	22	**30**	2	4	4	4	8	8	**30**
IV	CO4	14	**19**	4	4	4	4	4		**20**
V	CO5	10	**13**	2		4	4		4	**14**
	Total	**70**	**94**	**14**	**16**	**16**	**16**	**16**	**16**	**94**

Table For V Units:

Unit No.	I							II							III							IV							V							Total
CO	1							2							3							4							5							
Marks per Unit	14							10							22							14							10							70
1.35 Times marks	19							13							30							19							13							94
Bits	a	b	c	d	e	f	g	a	b	c	d	e	f	g	a	b	c	d	e	f	g	a	b	c	d	e	f	g	a	b	c	d	e	f	g	Total
CO	1	1	1	1	1	1	1	2	2	2	2	2	2	2	3	3	3	3	3	3	3	4	4	4	4	4	4	4	5	5	5	5	5	5	5	
Q1	2								2	2								2								2	2								2	14
Q2	4								4								4								4											16
Q3	4															4								4								4				16
Q4	4															4								4								4				16
Q5	4															4	4								4											16
Q6								4								4	4															4				16
Sub Total	18							12							30							20							14							94
TOTAL																																				

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